

AI511/MM505 Linear Algebra with Applications

Lecture 7 – Projections (cont.), Compositions

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September 29, 2025

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Outline

Recap

Euclidean vector spaces (cont.)

Examples of linear transformations (cont.)

Compositions of linear transformations

Recap

The standard basis

- The set of vectors $\{e_1, e_2, \dots, e_n\}$, where

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$$

is called the *standard basis* in $\mathbb{R}^n$.

- Any vector $v \in \mathbb{R}^n$ can be written in terms of the vectors $e_1, e_2, \dots, e_n$ in the following way

$$v = v_1 e_1 + v_2 e_2 + \dots + v_n e_n. \tag{1}$$

Matrix operators or matrix transformations

Every matrix $A \in \mathbb{R}^{m \times n}$ corresponds to a *matrix operator* or a *matrix transformation* $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$T_A(x) := Ax$$

for $x \in \mathbb{R}^n$.

Linear transformation

Definition 1.4.1 of Johnston (2021)

A *linear transformation* is a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ that satisfies the following two properties:

- (a) $T(v + w) = T(v) + T(w)$ for all vectors $v, w \in \mathbb{R}^n$;
- (b) $T(cv) = cT(v)$ for all vectors $v \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$.

Visually speaking, a transformation is “linear” if it has two properties

- (i) all lines remain lines, without getting curved;
- (ii) the origin remains fixed in place.

Matrices as linear transformations

Theorem 1.4.1 of Johnston (2021)

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if and only if there exists a matrix $[T] \in \mathbb{R}^{m \times n}$ such that

$$T(v) = [T]v \quad \text{for all } v \in \mathbb{R}^n.$$

Furthermore, the unique matrix $[T]$ with this property is called the *standard matrix* of T , and it is

$$[T] := \begin{bmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{bmatrix}.$$

In simple terms, a transformation is linear if and only if it is a matrix transformation.

Euclidean vector spaces (cont.)

Euclidean vector spaces (cont.)

Examples of linear transformations (cont.)

Orthogonal projection onto the direction of a unit vector u

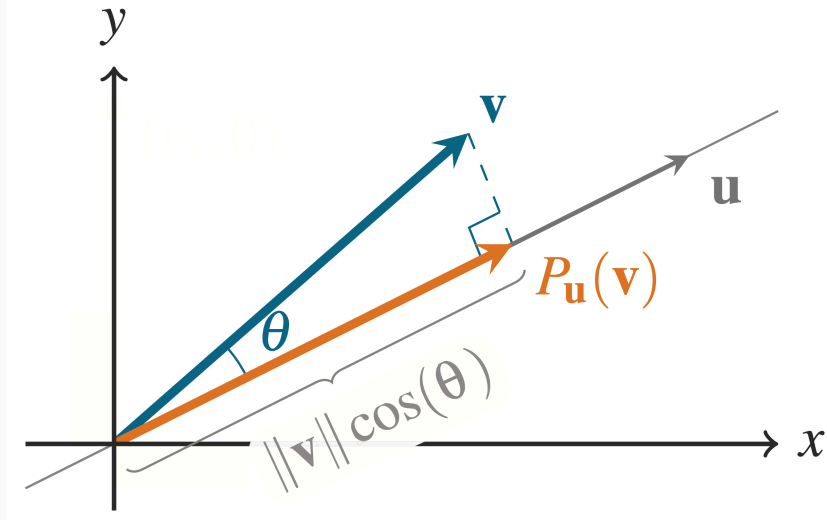


Figure 1.16 of Johnston (2021)

Orthogonal projection onto the direction of a unit vector u

- The projection of a vector v onto the line in the direction of u is denoted by $P_u(v)$, and it points a distance of $\|v\| \cos(\theta)$ (might be negative) in the direction of u , where θ is the angle between v and u .
- For $v, w \in \mathbb{R}^2$, we have that

$$\cos(\theta) = \frac{v \cdot w}{\|v\| \|w\|},$$

$$P_u(v) = \|v\| \cos(\theta) u = \|v\| \frac{(v \cdot u)}{\|v\| \|u\|} u = uu^T v$$

using the fact that $\|u\| = 1$ and $v \cdot u = u^T v$.

- Hence, $P_u(v)$ is a linear transformation, and its standard matrix is given by

$$[P_u] = uu^T \in \mathbb{R}^{n \times n}.$$

Orthogonal projection onto the direction of a vector w

In general, we have that

$$P_w(v) = \frac{1}{\|w\|^2} ww^T v = \frac{ww^T}{w^T w} v$$

for two vectors $v, w \in \mathbb{R}^n$ provided that $w \neq 0$.

Euclidean vector spaces (cont.)

Compositions of linear transformations

Composition of linear transformations

Definition 1.4.2 of Johnston (2021)

Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations. Then *composition* of S and T is the function $S \circ T : \mathbb{R}^n \rightarrow \mathbb{R}^p$ defined by

$$(S \circ T)(v) = S(T(v)) \quad \text{for all } v \in \mathbb{R}^n.$$

- T sends $\mathbb{R}^n$ to $\mathbb{R}^m$ and S sends $\mathbb{R}^m$ to $\mathbb{R}^p$ while the composition $S \circ T$ skips the intermediate step and sends $\mathbb{R}^n$ directly to $\mathbb{R}^p$.
- Is $S \circ T$ linear? If so, what is its standard matrix?

Composition is linear

Theorem 1.4.2 of Johnston (2021)

Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations with standard matrices $[T] \in \mathbb{R}^{m \times n}$ and $[S] \in \mathbb{R}^{p \times m}$, respectively. Then $S \circ T : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is a linear transformation, and its standard matrix is

$$[S \circ T] = [S][T].$$

- In simple terms, composition of linear maps is a matrix product of standard matrices.
- This theorem is the main reason that matrix multiplication was defined in the seemingly bizarre way.
- Recommended to watch *Matrix multiplication as composition* of 3Blue1Brown.

Example on the blackboard

- Consider two linear transformations

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix}.$$

- The linear transformation B
 - (i) rotates e_1 counterclockwise by 45° and extends its length to $\sqrt{2}$;
 - (ii) rotates e_2 counterclockwise by 90° and extends its length to 2.
- The linear transformation A
 - (i) flips e_1 and e_2 around 45° line;
 - (ii) extends the length of e_2 to 2 (stretches the space horizontally by 2).
- Find the standard matrix of the composition AB by describing where e_1 and e_2 are mapped and confirm your result by multiplying A and B .



Johnston, Nathaniel (2021). *Introduction to linear and matrix algebra*. Springer Cham.